

Validity Analysis: GSM8K

Target context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Overall risk: **HIGH**

Deployment Context

We are a US edtech company building an auto-grading system for American schools. Teachers assign arithmetic word problems and students submit numerical answers through our text-based platform. Our LLM independently solves the same problem and compares its answer to the student's submission. We need to evaluate whether the LLM reliably produces correct final numerical answers to arithmetic word problems.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	2	Significant gaps	high
Input Content ✓	4	Minor gaps	high
Input Form ✓	5	Strong alignment	high
Output Ontology ■	2	Significant gaps	high
Output Content	3	Moderate gaps	medium
Output Form	3	Moderate gaps	high
Average	3.2		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GSM8K is a partially valid evaluation for the US K-8 Arithmetic Auto-Grading Platform's deployment context. It is strongly aligned on input form, input content, and the basic chain-of-thought output pattern, all of which match the platform's text-only English, US-contextualized infrastructure. However, the two HIGH-priority dimensions (input ontology, output ontology) both show significant gaps: (1) GSM8K's problem-type coverage omits user-confirmed essential categories — standalone unit conversion, basic geometry (perimeter/area), and tabular/chart data interpretation — and under-represents grade 3–4 single-step difficulty; (2) GSM8K's exact-match scoring does not stress-test the answer-equivalence and tolerance-based normalization the platform's grading logic must handle. The benchmark provides a useful but incomplete reliability signal — necessary for assessing multi-step arithmetic reasoning in grades 5–8 but insufficient for the platform's full problem

distribution and grading requirements. Saturation on frontier LLMs further reduces its discriminative value.

Practical Guidance

What This Benchmark Measures

GSM8K can reliably evaluate an LLM's ability to solve US-contextualized multi-step arithmetic word problems requiring 2–8 reasoning steps, particularly in the grades 5–8 difficulty range. Its strongest dimensions (input_form: 5, input_content: 4) make it well-suited as a baseline reliability signal for the platform's English, text-only, US-dollar-and-imperial-unit problem profile. For multi-step ratio, percent, rate, and proportional reasoning problems [DATASET-D15, D16, D18, D20], GSM8K provides a representative evaluation.

Construct Depth

The benchmark probes final-answer correctness on multi-step arithmetic but does not probe at depth: (a) problem-type coverage is bounded to narrative arithmetic — omitting geometry, standalone unit conversion, and data interpretation (input_ontology: 2); (b) scoring is exact-match without equivalence normalization, so it does not probe whether the LLM handles representational variants the platform must grade (output_ontology: 2); (c) saturation on frontier models means a high score reflects coverage of a solved sub-problem rather than evidence of full reliability. The benchmark provides surface-level coverage of multi-step reasoning rather than a comprehensive reliability probe.

What Else You Need

To complete the assessment, the platform should supplement GSM8K with: (1) ASDiv-W [WEB-7] for explicit Geometry and UnitTrans problem-type coverage, addressing the input_ontology gap; (2) GSM-Plus integer-decimal-fraction conversion subset [WEB-9] for stress-testing answer-equivalence handling, addressing the output_ontology gap; (3) bespoke evaluation items authored by US K-8 educators for data interpretation from tables/bar charts, which no existing benchmark covers (input_ontology / input_form gap); (4) GSM1K [WEB-15] for contamination-controlled multi-step arithmetic evaluation given GSM8K saturation; (5) Easy2Hard-Bench / E2H-GSM8K [WEB-10] for difficulty-stratified evaluation approximating grade-band breakdown, addressing the grade 3–4 calibration gap; (6) sample re-annotation by US K-8 educators on edge cases to address output_content demographic uncertainty.

ID	Evidence
D15	Bobby has 16 toy cars, and the number of cars he has increases by 50% every year. How many toy cars will Bobby have in three years? — Compound percentage growth over multiple years — grades 6–8 level
D16	Last year Jessica paid \$1000 for rent, \$200 for food, and \$100 for car insurance each month. This year her rent goes up by 30%, food costs increase by 50%... — Multi-step percent change problem — upper grade band
D18	Ariella's account earns her simple interest at the rate of 10% per annum. — Simple interest calculation — finance context, grades 7–8
D20	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends $\frac{3}{4}$ of what she has on toys, how many will she remain with? — Ratio + fraction multi-step — upper grade band difficulty
WEB-7	ASDiv explicitly includes 'Geometry' and 'UnitTrans' problem categories that GSM8K lacks (https://arxiv.org/abs/2106.15772)
WEB-9	GSM-Plus tests robustness to answer-format perturbations not covered by GSM8K's exact-match scoring (https://arxiv.org/abs/2402.19255)
WEB-10	https://arxiv.org/pdf/2510.18147
WEB-15	https://arxiv.org/pdf/2405.00332

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input ontology is the HIGH-priority dimension for this deployment, and GSM8K shows significant gaps. The benchmark's scope is explicitly limited to multi-step grade-school arithmetic word problems [Q9, Q11], with no systematic subtask taxonomy by problem type. Three user-confirmed essential problem types — measurement/unit conversion, basic geometry (perimeter/area), and data interpretation from tables/bar charts — are absent or only incidentally present. The dataset analysis confirms zero standalone unit conversion problems (conversion appears only as embedded sub-steps in D7, D8) and zero geometry formula problems (D9 uses area as input, D10 computes prism volume incidentally). Data interpretation from tables/charts is structurally impossible in this text-only benchmark. Additionally, the difficulty distribution skews to 2–8 step problems [WEB-4, WEB-5], under-representing grade 3–4 single-step arithmetic (DATASET-D17, D18, D19 show the lower-bound complexity is still upper-grade-band).

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
D7	Roberto can skip 4,200 times an hour. Valerie can skip 80 times a minute. If they jump rope for fifteen minutes straight, how many skips will they total? — Implicit unit conversion (hours-to-minutes) embedded within a rate problem
D8	Larry spends half an hour twice a day walking and playing with his dog. He also spends a fifth of an hour every day feeding his dog. How many minutes does Larry spend on his dog each day? — Explicit hours-to-minutes conversion embedded in problem
D9	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area used as input quantity, but no geometry formula required — pure multiplication
D10	First calculate the volume of the aquarium by multiplying its length, width and height: $4\text{ ft} \times 6\text{ ft} \times 3\text{ ft} = 72\text{ cubic ft}$ — Volume of rectangular prism computed — closest geometry-adjacent example in sample
D17	Jim has 4 times as many Buicks as Fords, and 3 more than twice the number of Fords than Chevys. — Algebraic system-of-equations approach — above typical grade 3–4
D18	Ariella's account earns her simple interest at the rate of 10% per annum. — Simple interest calculation — finance context, grades 7–8
D19	Ten friends decide to get an end-of-year gift for their teacher. They plan to split the cost of the gift equally. But four of the group drop out. — Algebraic equation set-up ($10N = 6(N+8)$)
WEB-4	GSM8K problems each require between 2 and 8 solution steps; calibrated toward bright middle-school student (https://www.emergentmind.com/topics/gsm8k-dataset)
WEB-5	https://klu.ai/glossary/GSM8K-eval

Strengths

- Strong coverage of multi-step ratio, percent, rate, and proportional reasoning aligned to grades 6–8 [DATASET-D15, D16, D18, D20]
- Linguistic diversity within the arithmetic domain is an explicit design goal [Q9], reducing template-driven overfitting risk
- Explicit positioning as grade-school level with elementary concepts [Q11] aligns with the broad deployment domain

ID	Evidence
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Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
D15	Bobby has 16 toy cars, and the number of cars he has increases by 50% every year. How many toy cars will Bobby have in three years? — Compound percentage growth over multiple years — grades 6–8 level
D16	Last year Jessica paid \$1000 for rent, \$200 for food, and \$100 for car insurance each month. This year her rent goes up by 30%, food costs increase by 50%... — Multi-step percent change problem — upper grade band
D18	Ariella's account earns her simple interest at the rate of 10% per annum. — Simple interest calculation — finance context, grades 7–8
D20	The ratio of coins that Elsa has to that which Amalie has is 10:45. If the total number of coins they have is 440, and Amalie spends 3/4 of what she has on toys, how many will she remain with? — Ratio + fraction multi-step — upper grade band difficulty

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories per elicitation: single-step multiplication/division (grades 3–4), multi-step arithmetic, ratio/proportion, percent, rate, measurement/unit conversion, time, basic geometry (perimeter/area), and simple data interpretation from tables/bar charts.

IO-2: Confirmed omissions: (a) standalone unit conversion — only embedded incidentally [DATASET-D7, D8]; (b) geometry formula application — absent (DATASET-D9, D10 nearest, neither computes perimeter/area); (c) tabular/chart data interpretation — structurally impossible in text-only format; (d) single-step grade 3–4 calibration — paper acknowledges 175B model needs 2 orders of magnitude more data to reach 80% [Q50], and minimum sampled complexity is 2-step [DATASET-D29, D30].

IO-3: No clearly irrelevant categories identified; all problem types in GSM8K (multi-step arithmetic, ratio, percent, rate) are within the platform's stated scope.

IO-4: Documented gaps: unit conversion (HIGH), geometry (HIGH), data interpretation (HIGH/structural), grade 3–4 single-step calibration (MODERATE). These harm content validity because the benchmark systematically under-represents categories the platform actually serves.

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q11	"At the same time, GSM8K solutions depend only on elementary concepts, so achieving high test performance is a tractable goal." (p.2)
Q50	"Nevertheless, it appears likely that the 175B model would require at least two additional orders of magnitude of training data to reach an 80% solve rate." (p.6)
D7	Roberto can skip 4,200 times an hour. Valerie can skip 80 times a minute. If they jump rope for fifteen minutes straight, how many skips will they total? — Implicit unit conversion (hours-to-minutes) embedded within a rate problem
D8	Larry spends half an hour twice a day walking and playing with his dog. He also spends a fifth of an hour every day feeding his dog. How many minutes does Larry spend on his dog each day? — Explicit hours-to-minutes conversion embedded in problem
D9	The price of a home is \$98 per square foot (sq ft). The house is 2,400 sq ft and the barn out back is 1,000 sq ft. How much is this property? — Area used as input quantity, but no geometry formula required — pure multiplication

D10	First calculate the volume of the aquarium by multiplying its length, width and height: $4\text{ ft} * 6\text{ ft} * 3\text{ ft} = <<4*6*3=72>>72$ cubic ft — Volume of rectangular prism computed — closest geometry-adjacent example in sample
D29	Kat decides she wants to start a boxing career. She gets a gym membership and spends 1 hour in the gym 3 times a week... She also trained at the boxing gym 4 times a week for 1.5 hours. — Simple 2-step problem — among the lower-complexity examples in sample
D30	Five coaster vans are used to transport students for their field trip. Each van carries 28 students, 60 of which are boys. How many are girls? — 2-step problem; simple subtraction after multiplication
WEB-1	https://www.thecorestandards.org/standards-in-your-state/
WEB-3	https://goblinsapp.com/standards/math
WEB-4	GSM8K problems each require between 2 and 8 solution steps; calibrated toward bright middle-school student (https://www.emergentmind.com/topics/gsm8k-dataset)
WEB-7	ASDiv explicitly includes 'Geometry' and 'UnitTrans' problem categories that GSM8K lacks (https://arxiv.org/abs/2106.15772)
WEB-8	No K-8-calibrated benchmark combining arithmetic word problems with table/chart inputs exists; TAT-QA is finance-domain (https://aclanthology.org/2021.acl-long.254/)

Information Gaps

- Statistical frequency of unit-conversion-like and geometry-adjacent problems in the full 8,500-item dataset (only 47 main examples sampled)
- Whether any test-split problems differ in type distribution from the train split observed

Requires Expert Verification

- Mapping of confirmed essential problem types to specific CCSS-M codes (e.g., 3.OA, 4.MD, 6.RP) for precise gap accounting
- Platform-internal data interpretation modality (text-serialized vs. image-rendered tables)

Remediation

Gap 1: Standalone unit conversion and basic geometry (perimeter/area) problem types are absent or only incidentally present, despite being confirmed essential for grades 3–8.

Recommendation: Supplement GSM8K with ASDiv-W [WEB-7], which provides explicit 'UnitTrans' and 'Geometry' problem categories with grade-level annotations; use grade-level metadata to filter for the platform's 3–8 range.

Gap 2: Data interpretation from tables/bar charts is structurally impossible to cover with any existing text-only K-8 arithmetic benchmark.

Recommendation: Author a bespoke evaluation set of 100–200 data-interpretation items matching actual teacher-assigned content (text-serialized tables or image-rendered charts depending on platform modality), validated by US K-8 educators.

Gap 3: Grade 3–4 single-step arithmetic is underrepresented; the minimum sampled complexity is 2 steps and many problems require algebraic or compound reasoning.

Recommendation: Use E2H-GSM8K [WEB-10] difficulty stratification to report accuracy by tier separately, and supplement with single-step problem sets from MAWPS or ASDiv-A subset filtered for grades 3–4.

ID	Evidence
WEB-7	ASDiv explicitly includes 'Geometry' and 'UnitTrans' problem categories that GSM8K lacks (https://arxiv.org/abs/2106.15772)
WEB-10	https://arxiv.org/pdf/2510.18147

Input Content (IC)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is a LOWER-priority dimension and is one of GSM8K's stronger fits. Dataset analysis confirms US dollar amounts, imperial units (pounds, feet, inches, cubic feet), American names, and familiar US school/sports/consumer scenarios across the sample [\[DATASET-D1, D2, D3, D4, D5\]](#). One metric instance ([DATASET-D11](#), kilograms) appears, broadly matching the elicitation's description of a 'small minority' of metric problems. Minor concerns: annotator demographics are not documented [\[Q103, Q104\]](#), and GPT-3-generated seed questions [\[Q110\]](#) partly shaped problem framing rather than curated by US educators. Cherry-picking for diversity in appendix samples [\[Q127\]](#) limits confidence about systematic representativeness, but the empirical evidence strongly supports cultural alignment.

ID	Evidence
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q110	"To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model." (p.15)
Q127	"Samples were slightly cherry-picked for diversity." (p.18)
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US consumer goods, dollar amounts — typical birthday party scenario
D2	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9. — Imperial weight units (pounds), US-style pet scenario
D3	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Imperial length units; volume calculation (length × width × height)
D4	If Sally makes \$6 per day and Bob makes \$4 per day, how much money will they both have saved for their trip after a year? — Dollar amounts, annual savings rate problem
D5	He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. — American football scoring scenario — culturally specific to US
D11	Four people lost a total of 103 kilograms of weight. — Metric units (kilograms) appear in at least one problem

Strengths

- US monetary contextualization is pervasive — dollars, cents, tax, savings scenarios [\[DATASET-D1, D4, D6, D27\]](#)
- Imperial units dominate with occasional metric, matching platform distribution [\[DATASET-D2, D3, D11\]](#)
- Linguistic diversity is an explicit design principle with template-reuse detection [\[Q9, Q113\]](#)
- American school, sports, and consumer scenarios match deployment profile [\[DATASET-D5, D1, D32\]](#)

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)

Q113	"To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors." (p.15)
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US consumer goods, dollar amounts — typical birthday party scenario
D2	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9. — Imperial weight units (pounds), US-style pet scenario
D3	If the aquarium is 4 feet long, 6 feet wide, and 3 feet high, how many cubic feet of water are in the aquarium? — Imperial length units; volume calculation (length × width × height)
D4	If Sally makes \$6 per day and Bob makes \$4 per day, how much money will they both have saved for their trip after a year? — Dollar amounts, annual savings rate problem
D5	He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. — American football scoring scenario — culturally specific to US
D6	Each robot costs \$8.75. He was charged \$7.22 total for tax. He left with \$11.53 in change. How much did Austin start with? — Dollar amounts with cents (decimal currency), tax scenario
D11	Four people lost a total of 103 kilograms of weight. — Metric units (kilograms) appear in at least one problem
D27	\$68.47 total spent in-store + \$11.53 in change = \$<68.47+11.53=80>>80 to start. — Decimal dollar amounts in intermediate steps; final answer is integer
D32	Each bird eats 12 beetles per day, each snake eats 3 birds per day, and each jaguar eats 5 snakes per day. If there are 6 jaguars in a forest, how many beetles are eaten each day? — Multi-step chain multiplication (rate problem); no cultural specificity

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Inputs require US-typical cultural knowledge — dollar pricing, American school scenarios, US sports (football scoring in [DATASET-D5](#)). This aligns with the deployment target.

IC-2: Cultural alignment is strong: dataset analysis confirms predominantly US contextualization [[DATASET-D1](#), [D2](#), [D4](#), [D5](#)] matching the elicitation's description of platform problems as 'almost entirely US-contextualized.'

IC-3: Minimal Western-non-US content. Metric units appear sporadically [[DATASET-D11](#)] but do not constitute a systematic mismatch given the elicitation acknowledges some metric problems exist in science-adjacent platform content.

IC-4: INSUFFICIENT DOCUMENTATION — annotator demographics are not reported beyond identifying Upwork and Surge AI [[Q103](#), [Q104](#)]. Would need demographic data on contractor pool to confirm US-schooled framing.

IC-5: Content issues are limited: undocumented annotator demographics and GPT-3-seeded generation [[Q110](#), [Q111](#)] introduce mild risk that some problem framings were not curated by US K-8 educators, but empirical evidence does not surface concrete misalignments.

ID	Evidence
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)

Q110	"To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model." (p.15)
Q111	"Contractors were allowed to use those seed questions directly, to use them as inspiration and make modifications, or to come up with their own questions entirely." (p.15)
D1	Her parents bought a unicorn piñata for \$13 and filled it with all of her favorite treats. They bought 4 bags of Reese's for \$9 per bag, 3 bags of Snickers for \$5 per bag, and 5 bags of Skittles for \$7 per bag. — US consumer goods, dollar amounts — typical birthday party scenario
D2	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9. — Imperial weight units (pounds), US-style pet scenario
D4	If Sally makes \$6 per day and Bob makes \$4 per day, how much money will they both have saved for their trip after a year? — Dollar amounts, annual savings rate problem
D5	He scores 4 touchdowns per game and each touchdown is worth 6 points. There are 15 games in the season. — American football scoring scenario — culturally specific to US
D11	Four people lost a total of 103 kilograms of weight. — Metric units (kilograms) appear in at least one problem

Information Gaps

- Annotator demographics (geographic location, native language, US-schooled status)
- Frequency of metric vs. imperial unit problems across the full 8,500-item dataset

Input Form (IF)

Score: 5/5 — Strong alignment | Confidence: high

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is a LOWER-priority dimension and aligns nearly perfectly. Both benchmark and platform are text-only, English-language, US-standard mathematical notation (**DATASET-D33** shows the consistent calculator-annotation format). No script, modality, or language mismatch exists. The calculator-annotation training format **[Q40]** and natural-language chain-of-thought solutions **[Q32, Q33]** are structurally congruent with the LLM-as-solver deployment pattern. The only minor caveat is that the socratic config **[DATASET-D31]** provides sub-question scaffolding that doesn't match deployment problem format, but this is a configuration choice the platform can avoid.

ID	Evidence
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q33	"Moreover, this choice enables our models to develop verbal analytical skills and to produce solutions that are more readily interpretable by humans." (p.5)
Q40	"To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set." (p.6)
D31	How many cubic feet are in the aquarium? ** First calculate the volume of the aquarium by multiplying its length, width and height — Socratic decomposition scaffolds sub-questions for each step
D33	Second person = 27 - 7 = <<27-7=20>>20 kg\n103 - 27 - 20 = <<103-27-20=56>>56 kg\n56/2 = <<56/2=28>>28 kg — Answer annotations show calculator-injection format `<<expr=result>>`

Strengths

- Text-only English format exactly matches platform infrastructure **[Q9]**
- Natural-language chain-of-thought solutions match LLM-as-solver deployment pattern **[Q32, Q33]**
- Calculator-annotation format provides clean step-by-step computation **[DATASET-D33]**
- No script or notation mismatch — Arabic numerals, standard US math notation throughout

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
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D33	Second person = 27 - 7 = <<27-7=20>>20 kg\n103 - 27 - 20 = <<103-27-20=56>>56 kg\n56/2 = <<56/2=28>>28 kg — Answer annotations show calculator-injection format `<<expr=result>>`

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target

IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match exactly: both are text-only English prose with US-standard mathematical notation [DATASET-D33].

IF-2: Regional infrastructure (US edtech platform) fully supports text-only LLM I/O; no capture-specification differences apply.

IF-3: Domain-specific form: chain-of-thought reasoning with calculator annotations [Q40, Q119] matches the platform's expected LLM solving pattern. One caveat: data interpretation problems may involve image-rendered tables in deployment, which GSM8K cannot represent — but this is captured under input ontology rather than form.

IF-4: No material form mismatches for the text-only problem subset. Modality gap for table/chart inputs is real but more appropriately categorized as an ontology gap (table reading is a problem type GSM8K lacks).

ID	Evidence
Q9	"We designed this dataset to have high linguistic diversity while relying on relatively simple grade school math concepts." (p.2)
Q32	"First, we focus attention on the space of natural language solutions, as this is a richer and more general solution format than pure mathematical expressions." (p.5)
Q40	"To mitigate this issue, we train all models to use a calculator by injecting calculation annotations into the training set." (p.6)
Q119	"During training, there is no special distinction between the annotated tokens and the rest of the solution: they are all just tokens." (p.17)
D33	Second person = $27 - 7 = 20$ kg $103 - 27 = 76$ kg $56/2 = 28$ kg — Answer annotations show calculator-injection format <code><<expr=result>></code>

Information Gaps

- Platform's exact rendering of data-interpretation problems (text-serialized vs. image-rendered tables) — affects whether any IF gap exists for that subset

Requires Expert Verification

- Platform team confirmation of chain-of-thought surfacing to students/teachers and table-rendering modality

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output ontology is the second HIGH-priority dimension and shows significant misalignment. GSM8K's correctness criterion reduces to a single extracted numerical string with binary correct/incorrect judgment based solely on final answer match [Q31, Q60]. The standard scoring pipeline strips currency symbols and units [WEB-6] but does not implement fraction-decimal equivalence, rounding tolerance, or mixed-number normalization — a structural mismatch with the platform's partial normalization logic that must handle representational equivalences (e.g., '1/2' vs. '0.5', '\$2.50' vs. '2.50'). GSM-Plus's findings [WEB-9] confirm this is a real LLM failure mode: all 25 tested models showed significant drops on integer-decimal-fraction perturbations. The dataset analysis surfaces conditions that generate these equivalences (decimal currency in DATASET-D6, D27; fractions in DATASET-D23, D24, D25; comma-formatted answers in DATASET-D21, D22), but the scoring schema doesn't stress-test them.

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
D6	Each robot costs \$8.75. He was charged \$7.22 total for tax. He left with \$11.53 in change. How much did Austin start with? — Dollar amounts with cents (decimal currency), tax scenario
D21	They will skip 2,250 times because $150 \times 15 = 2250$ — Final answer formatted as "2,250" (comma-formatted integer) — not a plain integer string
D22	$98 \times 3400 = 333200$ — Answer annotation contains "333200.00" (trailing decimal zeros); extracted answer "333200" — format normalization in scoring
D23	So the cost of one bottle of milk was $75/100 \times 12 = 9$. — Percent expressed as fraction (75/100) in reasoning — answer is integer
D24	Arllette is $3/4 \times 28$ years = 21 years old. — Fraction used in intermediate step; final answer is integer
D25	After giving away $4/5$ of the cupcakes, Anna has $60 / 5 = 12$ cupcakes — Fraction used in problem setup; intermediate and final answers are integers
D27	\$68.47 total spent in-store + \$11.53 in change = \$80 to start. — Decimal dollar amounts in intermediate steps; final answer is integer
WEB-6	GSM8K standard pipeline strips currency symbols and units but does not implement fraction-decimal equivalence, rounding tolerance, or mixed-number normalization (https://mbrenndoerfer.com/writing/gsm8k-evaluating-mathematical-reasoning-language-models)
WEB-9	GSM-Plus tests robustness to answer-format perturbations not covered by GSM8K's exact-match scoring (https://arxiv.org/abs/2402.19255)

Strengths

- Decision rule is mathematically objective — final answer correctness is unambiguous for integer answers [Q31, Q60]
- Token-level verifier provides interpretable per-token judgment [DATASET-D33] [Q72, Q73, Q147]
- Clean '##### N` answer format makes extraction unambiguous [DATASET-D13, D30]

ID	Evidence
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)
Q72	"We can either train verifiers to make a single scalar prediction conditioned on the entire generated solution, or to make a scalar prediction after each token in the solution." (p.9)
Q73	"By default, we choose the latter, training verifiers to make predictions after each token." (p.9)
Q147	"One benefit of the token-level verifiers is that these models become immediately interpretable: we can visualize the predicted value for each token and better understand how the verifier makes decisions on judging samples." (p.21)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — 4-step sequential problem; straightforward multi-step arithmetic
D30	Five coaster vans are used to transport students for their field trip. Each van carries 28 students, 60 of which are boys. How many are girls? — 2-step problem; simple subtraction after multiplication
D33	Second person = $27 - 7 = 20$ kg $103 - 27 - 20 = 56$ kg $56/2 = 28$ kg — Answer annotations show calculator-injection format ` <code><<expr=result>></code> `

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label is a single numerical string with binary correct/incorrect judgment [[Q31](#), [Q60](#)]. Categories are mathematically universal and regionally relevant.

OO-2: Missing categories for the deployment: no representation of answer equivalence forms (fraction $\leftrightarrow$ decimal, mixed $\leftrightarrow$ improper, currency formatting, rounding tolerance, unit-stripped vs. unit-included). The platform's grading logic needs these; the benchmark schema does not encode them.

OO-3: No categories encode non-regional values. The gap is structural (missing equivalence handling) rather than value-laden.

OO-4: Stakeholder-driven redesign would mean augmenting evaluation with equivalence-class scoring (e.g., GSM-Plus integer-decimal-fraction subset [[WEB-9](#)]) rather than redesigning core categories.

OO-5: Documented taxonomy issue: exact-match scoring [[Q29](#), [Q46](#)] without equivalence normalization is a structural validity gap — the benchmark's construct (final answer correctness) is operationalized more narrowly than the deployment requires. Confidence is high because both verbatim quotes and web sources confirm the scoring pipeline lacks equivalence logic.

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q31	"Verifiers are trained to judge the correctness of solutions, with the training signal determined solely by whether or not the solution reached the correct final answer." (p.5)
Q46	"Test performance is determined by a single low temperature ($T = 0$) sample for each test problem." (p.6)
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)

WEB-6	GSM8K standard pipeline strips currency symbols and units but does not implement fraction-decimal equivalence, rounding tolerance, or mixed-number normalization (https://mbrenndoerfer.com/writing/gsm8k-evaluating-mathematical-reasoning-language-models)
WEB-9	GSM-Plus tests robustness to answer-format perturbations not covered by GSM8K's exact-match scoring (https://arxiv.org/abs/2402.19255)

Information Gaps

- Frequency of non-integer final answers (fractions, mixed numbers) in the full test split — sample shows nearly all integers, suggesting low frequency

Requires Expert Verification

- Platform's exact tolerance thresholds and normalization logic for cross-comparison with GSM-Plus-style evaluation

Remediation

Gap 1: Exact-match scoring does not stress-test answer-equivalence (fractions↔decimals, currency formatting, mixed numbers, rounding tolerance) that the platform's grading logic must handle.

Recommendation: Add the GSM-Plus integer-decimal-fraction conversion perturbation subset [WEB-9] to the evaluation suite; additionally implement a custom equivalence-class scoring harness aligned to the platform's tolerance thresholds.

ID	Evidence
WEB-9	GSM-Plus tests robustness to answer-format perturbations not covered by GSM8K's exact-match scoring (https://arxiv.org/abs/2402.19255)

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output content is MODERATE priority. Label quality is reasonable for the core arithmetic types: dedicated re-solving QC found 1.7% disagreement [Q107], which authors estimate as the breaking-error rate, while acknowledging subtle errors may be more frequent [Q109]. The dataset analysis confirms clean integer labels with consistent `#### N` extraction [DATASET-D13, D30]. However, several concerns: (a) annotator demographics are not documented [Q103, Q104], so subtle convention mismatches (US currency formatting, imperial unit naming) cannot be ruled out; (b) calculator annotations are auto-generated, not human-validated, and have known imperfections [Q117, Q118] — DATASET-D2 shows a minor annotation inconsistency ('6+6' instead of '6*2'); (c) verifier-based scoring introduces additional label risk via false positives from flawed reasoning [Q61, Q155] and false negatives from ambiguous language [Q153, Q154]. For the platform's grading use case (LLM-as-solver, not verifier-as-grader), risk (c) is less directly relevant.

ID	Evidence
Q61	"In practice, some solutions will reach the correct final answer using flawed reasoning, leading to false positives." (p.7)
Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
Q117	"The calculator annotations were not provided by human contractors: they were generated by a combination of hard-coded logic and a finetuned language model." (p.17)
Q118	"The logic for auto-generating calculator annotations is imperfect. It is highly unlikely to generate any incorrect annotations, but it is not uncommon for it to ignore some lines that could be annotated." (p.17)
Q153	"The second row contains a problem where the solution is correct, but the verifier has rated it as incorrect. This is potentially due to the ambiguity between the "4 times" and the "4 potatoes" in the problem description." (p.22)
Q154	"The third row consists of another false negative example. However, unlike the previous example, here the model completion contains some faulty reasoning. As such, even though the final answer in the model completion was correct, the natural language explanation was incorrect, and so the verifier correctly assigned a low score." (p.22)
Q155	"The final row contains a false positive, where the model makes a mistake on the second step, where it subtracts 400 from the price of a diamond jewel instead of a gold one. Verifiers occasionally make mistakes with performing this variable binding of quantities to their relationships." (p.22)
D2	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9. — Imperial weight units (pounds), US-style pet scenario
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — 4-step sequential problem; straightforward multi-step arithmetic
D30	Five coaster vans are used to transport students for their field trip. Each van carries 28 students, 60 of which are boys. How many are girls? — 2-step problem; simple subtraction after multiplication

Strengths

- Dedicated re-solving QC pass with disagreement-driven repair/discard [Q105, Q106]

- Quantified residual error rate of 1.7% [Q107] provides a published quality floor
- Clean integer ground-truth labels for the large majority of problems [DATASET-D13, D30]
- Anti-template-reuse pairwise similarity check [Q113] improves label diversity

ID	Evidence
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q113	"To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors." (p.15)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — 4-step sequential problem; straightforward multi-step arithmetic
D30	Five coaster vans are used to transport students for their field trip. Each van carries 28 students, 60 of which are boys. How many are girls? — 2-step problem; simple subtraction after multiplication

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels for arithmetic problems are mathematically determined and reflect universal regional perspectives on numerical correctness. Risk concentrates at edge cases involving conventions (currency formatting, unit naming) where annotator demographics matter.

OC-2: Potential disagreement with US-schooled annotators is bounded by the documented 1.7% rate [Q107]; subtle errors may be higher [Q109]. No regional-population-specific re-annotation evidence available.

OC-3: INSUFFICIENT DOCUMENTATION — annotator demographics (age, geography, native language, educational background) are not reported beyond identifying Upwork and Surge AI as platforms [Q103, Q104]. Would need a Datasheets/Data Statements entry on contractor demographics.

OC-4: Re-annotation by US K-8 educators on a sample of problems would be a reasonable remediation, especially for any problems involving fractions, mixed numbers, or unit-stripped answers where convention matters.

OC-5: Aggregation: labels were validated via two-stage re-solving with disagreement repair/discard [Q105, Q106, Q107]. This conservative aggregation likely under-erases minority perspectives for objective math but could not surface culturally-specific convention disputes.

OC-6: Documented label issues: 1.7% breaking-error rate, plus potentially larger subtle-error rate [Q108, Q109]; calculator-annotation auto-generation imperfections [Q117, Q118] confirmed in DATASET-D2.

ID	Evidence
Q60	"Training solutions are labeled as correct or incorrect based solely on whether they reach the correct final answer." (p.7)

Q103	"We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork (upwork.com)." (p.15)
Q104	"We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection." (p.15)
Q105	"After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote." (p.15)
Q106	"We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded." (p.15)
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q108	"We estimate this to be the fraction of problems that contain breaking errors or ambiguities." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)
Q117	"The calculator annotations were not provided by human contractors: they were generated by a combination of hard-coded logic and a finetuned language model." (p.17)
Q118	"The logic for auto-generating calculator annotations is imperfect. It is highly unlikely to generate any incorrect annotations, but it is not uncommon for it to ignore some lines that could be annotated." (p.17)
D2	At 7 weeks old, the puppy weighed 6 pounds, but doubled in weight by week 9. — Imperial weight units (pounds), US-style pet scenario

Information Gaps

- Annotator demographics (geographic, educational, linguistic)
- Subtle-error rate (beyond the 1.7% breaking-error figure)

Requires Expert Verification

- Sample re-annotation by US K-8 educators on edge-case problems (fractions, mixed numbers, unit-stripped answers)

Remediation

Gap 1: Annotator demographics are undocumented, so subtle convention mismatches (US currency formatting, imperial unit naming) cannot be ruled out; residual subtle-error rate may exceed the documented 1.7% [Q107, Q109].

Recommendation: Conduct a focused re-annotation pass with US K-8 educators on a stratified sample of 200–500 problems, prioritizing problems with fractions, mixed numbers, or unit-stripped answers; quantify residual disagreement rate.

ID	Evidence
Q107	"We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors." (p.15)
Q109	"It is possible that a larger percentage of problems contain subtle errors." (p.15)

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: GSM8K | Context: US K-8 Arithmetic Auto-Grading Platform (Grades 3–8)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output form is MODERATE priority. The benchmark produces a final numerical answer extracted from a natural-language chain-of-thought solution [Q29, Q46], which structurally matches the deployment's LLM-as-solver pattern. The calculator-override mechanism [Q41, Q120] further aligns with reliable computation. However, two concerns: (1) the standard scoring pipeline does single-sample exact-match evaluation [Q29] without testing tolerance-based or equivalence-aware grading that the platform requires; (2) GSM8K saturation (>90% on frontier LLMs per [WEB-13]) reduces discriminative power for selecting among current models. The output format itself is well-matched; the gap is in what the scoring procedure evaluates rather than what the model produces.

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q41	"At test time, a calculator will override sampling when the model chooses to use these annotations." (p.6)
Q46	"Test performance is determined by a single low temperature (T = 0) sample for each test problem." (p.6)
Q120	"During testing, we override model sampling when a well-formatted annotation exists, specifically overwriting the token(s) directly following "=" and within <<. . . >>." (p.17)
WEB-13	Most frontier LLMs achieve >90% accuracy on GSM8K — discriminative power for frontier-model selection is reduced (https://www.emergentmind.com/topics/gsm8k)

Strengths

- Final numerical answer format matches deployment grading-pipeline input [Q29]
- Calculator-override mechanism reduces arithmetic-error noise in produced answers [Q41, Q120]
- Natural language chain-of-thought output matches LLM-as-solver deployment pattern [DATASET-D33]
- Clean `#### N` answer extraction convention is unambiguous [DATASET-D13, D30]

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
Q41	"At test time, a calculator will override sampling when the model chooses to use these annotations." (p.6)
Q120	"During testing, we override model sampling when a well-formatted annotation exists, specifically overwriting the token(s) directly following "=" and within <<. . . >>." (p.17)
D13	On Monday Buddy has 30 baseball cards. On Tuesday Buddy loses half of them. On Wednesday Buddy buys 12 baseball cards. On Thursday he buys a third of what he had on Tuesday. — 4-step sequential problem; straightforward multi-step arithmetic
D30	Five coaster vans are used to transport students for their field trip. Each van carries 28 students, 60 of which are boys. How many are girls? — 2-step problem; simple subtraction after multiplication
D33	Second person = 27 - 7 = <<27-7=20>>20 kg\n103 - 27 - 20 = <<103-27-20=56>>56 kg\n56/2 = <<56/2=28>>28 kg — Answer annotations show calculator-injection format `<<expr=result>>`

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Expected output modality (text with embedded final numerical answer) matches deployment exactly — both produce a numerical answer that the platform's grading logic then compares to student submission.

OF-2: Not applicable — no speech-based output requirement.

OF-3: Not applicable — platform users (grades 3–8 students) interact via text; LLM output is consumed by the grading pipeline, not the student.

OF-4: Form mismatches are minor: (a) exact-match scoring [Q29] doesn't exercise equivalence-aware grading the platform requires (overlaps with OO finding); (b) saturation [WEB-13] limits the benchmark's ability to discriminate among frontier models; (c) socratic config [DATASET-D31] would inflate scores if used — should use main config.

ID	Evidence
Q29	"At test time, we judge performance by autoregressively sampling a single low temperature solution and checking whether the final answer is correct." (p.5)
D31	How many cubic feet are in the aquarium? ** First calculate the volume of the aquarium by multiplying its length, width and height — Socratic decomposition scaffolds sub-questions for each step
WEB-9	GSM-Plus tests robustness to answer-format perturbations not covered by GSM8K's exact-match scoring (https://arxiv.org/abs/2402.19255)
WEB-13	Most frontier LLMs achieve >90% accuracy on GSM8K — discriminative power for frontier-model selection is reduced (https://www.emergentmind.com/topics/gsm8k)

Information Gaps

- Whether the platform's grading pipeline expects only the final number or full chain-of-thought; affects whether output-form match extends to intermediate reasoning

Requires Expert Verification

- Platform team confirmation of chain-of-thought consumption (final answer only vs. full reasoning surfaced to teachers/students)

Remediation

Gap 1: GSM8K saturation on frontier LLMs (>90% accuracy per [WEB-13]) limits discriminative power for selecting among current models the platform might deploy.

Recommendation: Use GSM1K [WEB-15] as a contamination-controlled complement to GSM8K, and rely on the supplementary benchmarks above for the differentiation that GSM8K alone cannot provide at the frontier.

ID	Evidence
WEB-13	Most frontier LLMs achieve >90% accuracy on GSM8K — discriminative power for frontier-model selection is reduced (https://www.emergentmind.com/topics/gsm8k)
WEB-15	https://arxiv.org/pdf/2405.00332